

ROTATIONAL MOTION

Rotational Motion

Definition:

The motion of a body about a fixed axis is called rotational motion.

Examples:

- Rotation of ceiling fan
- Motion of Earth about its axis
- Rotating wheel

Angular Displacement

Definition:

Angular displacement is the angle through which a body rotates about a fixed axis.

Symbol: θ **SI Unit:**

Radian (rad)

Angular Velocity**Formula:**

$$\omega = \theta / t$$

SI Unit:

rad/s

Angular Acceleration**Definition:**

Angular acceleration is the rate of change of angular velocity with time.

Formula:

$$\alpha = (\omega_2 - \omega_1) / t$$

SI Unit:

$$\text{rad/s}^2$$

Moment of Inertia**Definition:**

Moment of inertia is the rotational analogue of mass.

Formula:

$$I = mr^2$$

Where:

- I = Moment of inertia
- m = Mass of body
- r = Distance from axis

SI Unit:

$$\text{Kg}\cdot\text{m}^2$$

Radius of Gyration:

$$k = \sqrt{I / m}$$

Torque**Definition:**

Torque is the turning effect of a force acting on a body.

Formula:

$$\tau = rF \sin\theta$$

SI Unit:

Newton-meter (N·m)

Angular Momentum**Definition:**

Angular momentum is the product of moment of inertia and angular velocity.

Formula:

$$L = I\omega$$

SI Unit:

$\text{kg}\cdot\text{m}^2/\text{s}$

Conservation of Angular Momentum:

When no external torque acts on a body, its angular momentum remains constant.

Examples:

- Spinning ice skater
- Motion of planets
- Rotating fan

Applications of Rotational Motion:

- Motion of wheels
- Rotation of Earth
- Electric motors
- Ceiling fans
- Rolling objects

Rolling Motion:

Rolling motion is a combination of translational motion and rotational motion.

Example:

A rolling wheel on a road.

Quick Revision Points:

- Rotational motion occurs about a fixed axis.
- Angular displacement is represented by θ .
- Angular velocity is $\omega = \theta / t$.
- Moment of inertia is the rotational analogue of mass.
- Torque is the turning effect of force.
- Angular momentum is $L = I\omega$.
- Angular momentum remains conserved if no external torque acts.

Important Formulae:

- $\omega = \theta / t$
- $\alpha = (\omega_2 - \omega_1) / t$
- $I = mr^2$
- $\tau = rF \sin\theta$
- $L = I\omega$
- $k = \sqrt{I / m}$